Yibo Yin

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EDUCATION

École polytechnique fédérale de Lausanne (EPFL) — **Lausanne, Switzerland**
M.Sc. in Digital Humanities — *09/2025 - Present*

- Research Focus: Computer Graphics, Computer Vision, and Machine Learning
- **GPA:** 5.56 /6.00

Wuhan University — **Wuhan, China**
B.Eng. in Computer Science and Technology — *09/2021 - 06/2025*

- **GPA:** 3.93 /4.00 (top 4%)

PUBLICATIONS

MATStruct: High-Quality Medial Mesh Computation via Structure-aware Variational Optimization
Ningna Wang, Rui Xu, ***Yibo Yin***, Zichun Zhong, Taku Komura, Wenping Wang, Xiaohu Guo
ACM SIGGRAPH Asia 2025 (Conference Track)

RESEARCH EXPERIENCE

Image and Visual Representation Lab, EPFL — *06/2026 - Present*
Research Assistant | Advisor: Prof. Sabine Süsstrunk and Dr. Zhuoqian Yang

- Working on Weight Space Representation and Implicit Neural Representations.

Realistic Graphics Lab, EPFL — *03/2026 - 06/2026*
Research Assistant | Advisor: Prof. Wenzel Jakob

- Implemented differentiable Gaussian Process Implicit Surface (GPIS) sampling with Dr.Jit.
- Developed a hybrid surface – volume rendering approach for Many-Worlds inverse rendering.

Computer Graphics Lab, The University of Texas at Dallas — *04/2024 - 11/2025*
Research Assistant (Remote) | Advisor: Prof. Xiaohu Guo

- Extended the evaluation algorithm in the SIGGRAPH Course Note paper *Evaluation of Loop Subdivision Surfaces* and the SGP paper *Fitting Sharp Features with Loop Subdivision Surfaces* to structured medial axis meshes under Loop Subdivision.
- Extended the simplification algorithm in the SIGGRAPH paper *Q-MAT: Computing Medial Axis Transform by Quadratic Error Minimization* to structured medial axis meshes.
- Proposed a subdivision surface fitting algorithm in collaboration with advisor, designed to preserve the sharp features of the structured medial axis mesh during fitting.

Waterloo Computer Graphics Lab, University of Waterloo — *06/2024 - 08/2024*
Research Assistant (Remote) | Advisor: Prof. Toshiya Hachisuka

- Reproduced the algorithm in the SIGGRAPH paper *A Practical Walk-on-Boundary Method for Boundary Value Problems* for solving the boundary value problem of Laplace's equation with Dirichlet boundaries.

Graphics and Vision Lab, Wuhan University — *05/2023 - 01/2024*
Research Assistant | Advisor: Prof. Chunxia Xiao

- Assembled a set of equipment with an Intel depth camera and STMicroelectronics ToF sensors, creating a dataset of over 10,000 images and corresponding sensor data.
- Proposed a novel monocular depth estimation method using RGB images and sensor data as input.

SELECTED PROJECTS

Physically Based Renderer — *2026*
CS-440 Advanced Computer Graphics Rendering Competition, EPFL

- Extended the Nori framework with heterogeneous volumetric path tracing, delta and ratio tracking, Henyey-Greenstein phase functions, and MIS over emitter, phase, and BSDF sampling.
- Implemented HDR environment map lighting, thin-lens depth of field, motion blur, principled BSDF, and GGX rough dielectric materials for a storm-scene rendering pipeline.

AWARDS

- **Outstanding Student** (10% school-wide), Wuhan University — *2022, 2023, 2024*
- **Second Class Scholarship** (10% school-wide), Wuhan University — *2022*
- **Third Class Scholarship** (15% school-wide), Wuhan University — *2023, 2024*
- **Lei Jun Computer Innovation and Development Fund**, Wuhan University — *2024*

TECHNICAL SKILLS

- **Languages:** Mandarin (Native), English (TOEFL iBT 102)
- **Programming Languages:** C++, C, Python, C#, GLSL, SQL, Verilog HDL, Java
- **Libraries / Frameworks / Tools:** PyTorch, Dr.Jit, Mitsuba, PBRT-v3, Blender, Eigen, CGAL, libigl, Git, CMake